

SLM gray-level → phase calibration SOP

Why this exists, how to measure it for your panel and wavelength, and how to load the result into `slm_toolbox`. See [slm_toolbox/calibration.py](#) (curve fitting/storage) and [slm_toolbox/calibration_patterns.py](#) (the raw test patterns to display) for the code this feeds, and the "Calibration" panel in `phase_gui.py` for the UI — including a guided wizard for Method C below.

1. Why calibration is wavelength-dependent

A phase-only LC-SLM doesn't take "phase" as input — you send an 8-bit (or 16-bit) gray level per pixel. The panel's own driver converts that gray level to a voltage, which sets the liquid crystal's birefringence $\Delta n(V)$, which sets the phase retardation:

$$\delta = 2\pi \cdot d \cdot \Delta n(V) / \lambda$$

(d = LC cell thickness, λ = wavelength — standard birefringent-retarder physics.) For a fixed gray level (fixed voltage, fixed Δn), the resulting phase shift is inversely proportional to wavelength: the same 8-bit value gives you less phase shift at longer wavelengths. In practice Δn itself also has some wavelength dispersion, so it's not exactly $1/\lambda$, but $1/\lambda$ is the dominant, first-order effect — real datasheet numbers show the trend clearly on a single device, e.g. HOLOEYE's PLUTO-2.1-VIS-016 panel: 6.9π at 530nm vs. 5.2π at 633nm, same panel, same drive range [holoeeye.com/slm-pluto-phase-only]. This means a panel calibrated (gray 0-255 spans exactly 0- 2π) at one wavelength will not span exactly 0- 2π at another wavelength — it might only reach 1.5π , or it might wrap past 2π and become ambiguous. Calibration must be redone (or at least re-verified) per wavelength.

2. How HOLOEYE's own software handles this (for reference)

HOLOEYE's SLM Pattern Generator software's phase→gray-level algorithm explicitly assumes the panel is already calibrated so that gray 0-255 spans exactly 2.0π rad [holoeeye.com/products/spatial-light-modulators/slm-pattern-generator/] — the pattern-generation software itself does not adapt to wavelength; the calibration is applied at the device level. HOLOEYE panels (PLUTO-2.1, LUNA, LETO-3, GAEA-2.1, ERIS-1.1, GAEA-C) ship with a separate Configuration Manager utility that uploads "a new gamma curve or another digital drive scheme" onto the SLM controller itself via USB/virtual COM port [consistent across all listed HOLOEYE product pages]. In other words: HOLOEYE's answer to "different wavelength → different gamma" is swap the gamma curve loaded onto the device, done once per working wavelength via their vendor utility, after which the pattern-generation software's linear 0- 2π assumption is valid again.

We don't have that path — an HDMI-connected SLM has no vendor firmware access from a generic host; there's no equivalent to HOLOEYE's Configuration Manager here. `slm_toolbox` gets the same end result a different way: instead of correcting the device, `calibration.py` corrects the gray value we send (the `gamma_lut` passed into `render.phase_to_gray`), computed host-side from a measured curve for the current wavelength. Functionally equivalent output, done in software instead of firmware.

3. Measurement methods

Three practical options, roughly in order of rigor vs. equipment needed:

Method A — Common-path / self-referenced interferometry (no external interferometer needed, most rigorous)

The base pattern is wired up in the GUI: Calibration panel → "Self-Referenced Pattern (Method A)..." projects the grating-half/piston-half pattern below for whatever piston gray level you set — phase extraction from the resulting fringes is setup-specific (depends on your camera/optics), so that part isn't automated; once you have gray→phase points from your own fringe analysis, enter them via "Enter/Edit Calibration..."

Based on Martínez Fuentes et al., "Interferometric method for phase calibration in liquid crystal spatial light modulators using a self-generated diffraction-grating," Opt. Express 24(13):14159 (2016) — peer-reviewed, confirmed method:

- Split the SLM into two regions: one half displays a binary diffraction grating, the other half a uniform gray level (piston). The grating half diffracts light into a tilted plane wave; the piston half passes an undiffracted reference wave. These interfere directly — no separate interferometer arm, beamsplitter, or reference mirror needed; the SLM itself generates both waves. "Experimentally straightforward, robust, requires solely a collimated beam" [Opt. Express 24, 14159].
- Step the piston region's gray level through your full range (see §4 for sampling); a camera downstream of a small aperture/lens records the interference fringe pattern at each step. The measured fringe phase shift versus piston gray level is your gray→phase curve.
- This is the recommended method if you have a collimated beam and a camera but no dedicated interferometer optics.

Method B — Twyman-Green interferometry, SLM-driven phase stepping (most efficient, needs interferometer optics)

Based on Lee, Koo & Kim, "Simple and fast calibration method for phase-only spatial light modulators," Opt. Lett. 48(1):5-8 (2023) — peer-reviewed, confirmed method:

- Standard Twyman-Green interferometer, but instead of physically stepping a piezo-driven mirror to phase-shift the reference arm (conventional N-step phase-shifting interferometry), the SLM's own gray level modulates the interference intensity directly — no PZT needed.
- Notably faster: "requires N times fewer interferograms than typical N-step phase-shift interferometry" [Opt. Lett. 48, 5], and can simultaneously extract per-pixel phase-response nonuniformity and panel flatness nonuniformity, not just a single global curve — worth using if you already have interferometer optics on the bench and want a per-pixel-accurate map instead of one global curve.

Method C — Diffraction-efficiency-vs-gray-level (simplest, least equipment)

This method is fully wired up in the GUI: phase_gui.py's Calibration panel → "Calibration Wizard (Method C)..." steps you through it — pick a target monitor, it displays the reference-vs- test grating for each gray level, you type in the measured efficiency, and it computes + saves the calibration curve automatically once every level has a reading (via `slm_toolbox.calibration.unwrap_efficiency_sweep` — validated against synthetic ground-truth data, error shrinks with denser sampling as expected for this method). The practical fallback with just a laser, the SLM, and a power meter (or camera) — no interferometer at all:

- 1 Display a two-level binary phase grating on the SLM: half the pixels (e.g. alternating columns) at a fixed reference gray level g_{ref} (commonly 0), the other half at a test gray level g_{test} .
 - 2 Measure the 1st-order diffraction efficiency (power in the ± 1 diffracted order relative to incident power) as g_{test} is swept from 0 to 255.
 - 3 For a binary phase grating, 1st-order efficiency is proportional to $\sin^2(\Delta\phi/2)$ where $\Delta\phi$ is the phase difference between the two gray levels — efficiency is zero when $\Delta\phi = 0$ or 2π and maximum when $\Delta\phi = \pi$. Sweeping g_{test} and finding the efficiency minima/maximum gives you $\Delta\phi(g_{\text{test}})$ directly (invert the $\sin^2$ relation), without needing to measure phase directly.
 - 4 This is qualitatively less precise than A/B (no interferometric phase readout, and the $\sin^2$ inversion is ambiguous in sign/branch without extra care), but it's the right choice if you have no way to build an interference setup at all — it directly reuses the "grating diffraction efficiency" physics already in this project's forked-hologram work.
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4. Concrete procedure (applies to all three methods)

- 1 Sample enough gray levels. At minimum ~16-17 evenly spaced points across 0-255 (steps of 16); 33 points (steps of 8) is a safer default if your panel might be non-monotonic or have a flat (saturated) region near the ends. Always include the endpoints 0 and 255.
 - 2 Let the panel settle before each measurement — LC response has a settling time (ms-scale) and panel heating drifts the calibration slowly over the session; if you're doing a careful measurement, re-check a couple of reference points at the end and confirm they haven't drifted.
 - 3 Watch for flicker. Many LC-SLMs use an AC drive scheme with a flicker component; average over multiple frames per gray level rather than a single snapshot to avoid aliasing it into your phase measurement.
 - 4 Watch for non-monotonicity and saturation. Real panels often flatten out (phase stops increasing with gray level) near gray=0 and gray=max as the LC director hits its physical extremes.
`CalibrationCurve.to_gamma_lut` in this codebase already handles flat/non-increasing regions defensively (nudges them to be strictly increasing before inverting), but it's still best practice to identify and note where your panel saturates and avoid relying on that region.
 - 5 Confirm total span. After measuring, check whether your curve's phase actually reaches 2π within gray 0-255. If it falls short (e.g. only 1.5π), gray 255 alone can't give you a full 2π wrap — you're limited to whatever phase range you actually measured, which matters for patterns needing the full range (e.g. blazed gratings need a clean 2π sawtooth for high diffraction efficiency).
 - 6 Record wavelength precisely. The whole point of this SOP is that the curve is only valid at the wavelength you measured it at (§1) — always label your saved calibration with the exact λ used.
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5. Loading the result into `slm_toolbox`

Three ways, all in `phase_gui.py`'s "Calibration" panel:

- "Calibration Wizard (Method C)..." — computes and saves it for you as you go (§3, Method C) — the recommended path if you don't have interferometer optics.
- "Enter/Edit Calibration..." — paste your measured points as `gray,phase_rad` pairs (one per line) for the wavelength currently set in the Wavelength field, then Save. Use this for Method A/B results (phase from your own fringe analysis) or any other source. This writes `calibrations/<wavelength>nm.json`.

- "Load Calibration File..." — import a previously exported/saved JSON curve (same format — see `CalibrationCurve.to_dict/from_dict` in `slm_toolbox/calibration.py`).

Once saved, the GUI automatically resolves and applies the right curve whenever you Generate Preview / Save / Project, based on the Wavelength field:

- Exact match (within $\pm 2\text{nm}$ of a saved calibration) → uses it directly.
- No exact match, but another wavelength is calibrated → approximates via the $1/\lambda$ relation (`CalibrationCurve.rescale_to_wavelength`) and clearly labels the result as approximate. Treat this as a starting point, not a substitute for measuring at your actual working wavelength — §1's real HOLOEYE datasheet numbers show the scaling isn't perfectly $1/\lambda$ in practice (different LC mixtures have their own birefringence dispersion), so the further your working wavelength is from the measured one, the less accurate the rescale will be.
- Nothing calibrated at all → falls back to an explicit, clearly-labeled uncalibrated linear mapping (gray 0-255 → phase 0- 2π directly). This is almost certainly wrong on real hardware — it exists only so the software has something to render before you've calibrated, never as a claim of correctness.

Sources

- Physics: $\delta = 2\pi \cdot d \cdot \Delta n / \lambda$, standard LC birefringent-retarder phase retardation (textbook LC optics).
- HOLOEYE phase-depth-vs-wavelength datasheet figures: holoeeye.com/slm-pluto-phase-only/, holoeeye.com/luna-phase-only-spatial-light-modulator/.
- HOLOEYE Configuration Manager / gamma-curve-per-device architecture: consistent across HOLOEYE PLUTO-2.1, LUNA, LETO-3, GAEA-2.1, ERIS-1.1, GAEA-C product pages (holoeeye.com/products/spatial-light-modulators/).
- HOLOEYE SLM Pattern Generator's linear 0- 2π assumption: holoeeye.com/products/spatial-light-modulators/slm-pattern-generator/.
- Method A: Martínez Fuentes, Fernández, Prieto, Artal, "Interferometric method for phase calibration in liquid crystal spatial light modulators using a self-generated diffraction-grating," *Optics Express* 24(13):14159–14171 (2016).
- Method B: Lee, Koo & Kim, "Simple and fast calibration method for phase-only spatial light modulators," *Optics Letters* 48(1):5–8 (2023).
- Background review: "Progress in Phase Calibration for Liquid Crystal Spatial Light Modulators," *Applied Sciences* 9(10):2012 (2019) — covers both interferometric and diffractive/self-generated calibration method families.

Caveats (from adversarial fact-checking during research for this doc): cross-device comparisons of phase depth vs. wavelength (comparing different SLM product variants at different wavelengths) were found to overstate how cleanly $1/\lambda$ scaling holds, since different variants use different LC mixtures each optimized for their band — the same-device datapoints cited above (e.g. PLUTO-2.1-VIS-016 at two wavelengths) are more trustworthy evidence of the trend than cross-model comparisons. One secondary source disagreed with HOLOEYE's own LUNA-TELCO-115 spec (2.6π vs. 2.3π at 1550nm) — treat exact datasheet numbers as illustrative, not as a substitute for measuring your specific panel.